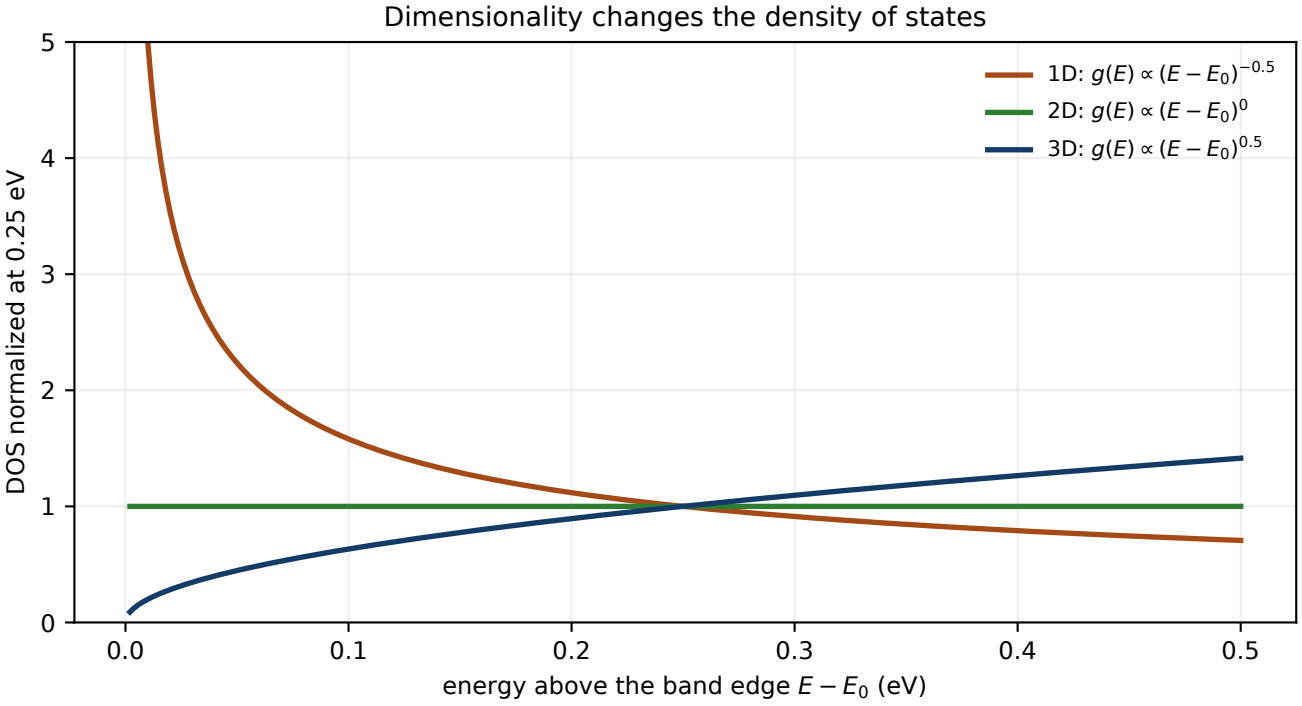
Dimensionality changes the density of states
1D: $g(E) \propto (E - E_0)^{-0.5}$
2D: $g(E) \propto (E - E_0)^0$
3D: $g(E) \propto (E - E_0)^{0.5}$
DOS normalized at 0.25 eV
energy above the band edge $E - E_0$ (eV)
0
1
2
3
4
5
0.0
0.1
0.2
0.3
0.4
0.5